

Part A — Review of Week 2 Analysis and Week 3 Improvement Plan

1. Review of Week 2 Analysis

A rigorous review of the Week 2 deliverables was conducted to assess the quality, completeness, and business value of the analysis completed so far.

Week 2 successfully established the cleaned datasets, foundational data-quality checks, initial reporting, and a core Power BI dashboard. These deliverables provided a strong analytical foundation for further development.

However, the review identified several opportunities to deepen the analysis and increase its business impact in Week 3. Three key areas were identified for advanced technical expansion

2. Area One: Granular Customer Segmentation and Dashboard Interactivity

Issue/Gap

Although the Week 2 Power BI dashboard effectively communicated overall transaction trends and performance, its customer segmentation and filtering capabilities remained relatively high-level.

Evidence

Stakeholders are currently unable to explore performance dynamically across more detailed dimensions, such as:

- Customer behavioral tiers

- Specific payment channels

- Transaction types

- Risk-review classifications

- Other relevant customer and transaction characteristics

This limits the ability to move from general reporting to more targeted customer and transaction analysis.

Required Improvement

In Week 3, the Power BI dashboard will be enhanced with:

- Advanced and interactive slicers

- More granular customer segmentation

- A dedicated risk-review analysis panel

- Deeper customer behavior drill-downs

- Interactive filtering across key transaction dimensions

The objective is to make the dashboard more exploratory and allow stakeholders to identify specific patterns and risk areas rather than relying only on aggregate results.

3. Area Two: Deepening Insights Through Advanced SQL Workflows

Issue/Gap

The Week 2 analysis relied substantially on flat-file data cleaning, spreadsheet-based checks, and foundational data exploration.

Evidence

The initial workflow did not fully utilize the analytical possibilities offered by a relational database. In particular, opportunities to dynamically cross-reference information across multiple tables were underexplored.

For example, customer profiles could be combined with transactional records to investigate relationships between customer characteristics, transaction behavior, payment channels, and transaction outcomes.

Required Improvement

Week 3 will introduce an advanced SQL analysis workflow consisting of at least eight complex queries.

The SQL analysis will incorporate:

- JOINS to combine related datasets and tables

- Common Table Expressions (CTEs) for structured and reusable analysis

- Window functions for ranking, comparisons, and trend analysis

- CASE statements for conditional classification and segmentation

The objective is to move beyond basic data retrieval and use SQL to uncover deeper transactional patterns and relationships that may not be immediately visible in the dashboard.

4. Area Three: Automated Statistical Validation Using Python

Issue/Gap

The Week 2 validation process depended primarily on manual spreadsheet sorting, basic data checks, and visual inspection.

Evidence

Programmatic validation and automated statistical profiling were not yet integrated into the analytical workflow. This created an opportunity to improve the reliability, efficiency, and reproducibility of the analysis.

Required Improvement

A dedicated Google Colab Python notebook will be introduced in Week 3 for automated data validation and statistical analysis.

The workflow will utilize:

- Pandas for data manipulation, cleaning, and profiling

- NumPy for numerical and statistical operations

- Seaborn for advanced statistical visualization

The notebook will include automated data profiling and generate at least five advanced statistical visualizations to investigate trends, distributions, relationships, and potential anomalies within the dataset.

The goal is to complement the Power BI and SQL analysis with a reproducible Python-based validation layer.

5. Week 3 Overall Objective

The primary objective of Week 3 is to transform the Week 2 foundation into a more interactive, technically robust, and insight-driven analytical workflow.

The three improvement areas will work together:

Power BI → Interactive business reporting

SQL → Deeper transactional analysis

Python → Automated statistical validation

Together, these enhancements will provide a more comprehensive view of customer behavior, transaction performance, and risk-related patterns while demonstrating practical proficiency across multiple data-analysis tools.